

## How Streamlit Communicates with FastAPI

These notes explain the requests library and the send\_message() function used to connect a Streamlit frontend with a FastAPI backend.

### 1. Import Requests Library

```
import requests
```

The requests library allows Python to send HTTP requests to web servers or APIs.

### 2. POST Request

```
response = requests.post(  
    "http://127.0.0.1:8000/chat",  
    json={"message": "Hello"}  
)
```

POST is used to send data to the FastAPI server.

### 3. Response Object

```
print(response)
```

Returns something like: <Response [200]>

### 4. Status Code

```
response.status_code
```

200 = Success, 404 = Not Found, 500 = Server Error

### 5. JSON Response

```
response.json()
```

Converts the JSON returned by FastAPI into a Python dictionary.

### 6. Access Dictionary Value

```
response.json()["answer"]
```

Returns only the value stored in the "answer" key.

### 7. f-String

```
BASE_URL = "http://127.0.0.1:8000"  
url = f"{BASE_URL}/chat"
```

f-strings insert variable values into strings.

## 8. JSON Parameter

```
json={
    "message": message
}
```

Python automatically converts the dictionary into JSON before sending it.

## 9. Complete Function

```
def send_message(message):

    response = requests.post(
        f"{BASE_URL}/chat",
        json={
            "message": message
        }
    )

    return response.json()["answer"]
```

## 10. Line-by-Line Explanation

1. Create a function called `send_message()`.
2. Send an HTTP POST request to FastAPI.
3. Build the URL using an f-string.
4. Send the user's message as JSON.
5. Receive the server response.
6. Convert the response to a Python dictionary.
7. Return only the value of the 'answer' key.

## 11. Complete Communication Flow

```
graph TD
    User --> StreamlitUI[Streamlit UI]
    StreamlitUI --> requests[requests.post()]
    requests --> FastAPI[FastAPI Server]
    FastAPI --> LLM[LLM (Groq/OpenAI/etc.)]
    LLM --> JSONResponse[JSON Response]
    JSONResponse --> response[response.json()]
    response --> Display[Display Answer in Streamlit]
```

## 12. Example

```
question = "What is FastAPI?"  
answer = send_message(question)  
print(answer)
```